

Data-Driven Texas Livestock Markets

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Meet the Team



Stiles Clements



Emma Trunnell



Keenan Hui



Abhiroop Kumar



Suryah Vadivel

The Sponsor



- Founded in 1919, Columbus State Bank is a community bank in Colorado County, TX
- Services: checking, savings, IRA, and agricultural/commercial loans



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— STATE BANK —

The Problem

How can we characterize the Texas livestock auction market to identify which segments represent the greatest opportunity for modernizing the industry's largely manual payment system?

The Market

~180 licensed Auction Barns Across Texas

- Barns have weekly livestock sales
- Settlements are entirely manual and check-based
- The ecosystem is fragmented, trust-driven, and largely un-digitized.

The Business Opportunity

- Modernize payment flows for faster, transparent transactions
- Automate reconciliation and compliance tracking
- Position CSB to capture value from transaction flows

Dataset

915 weekly reports (Jan 2024 - Mar 2026) across 44 barns

Seven-step cleaning pipeline, including:

- Standardized pricing (\$/cwt, \$/lb, \$/head)
- Removal of outliers, inconsistencies, duplicates

Core analysis set

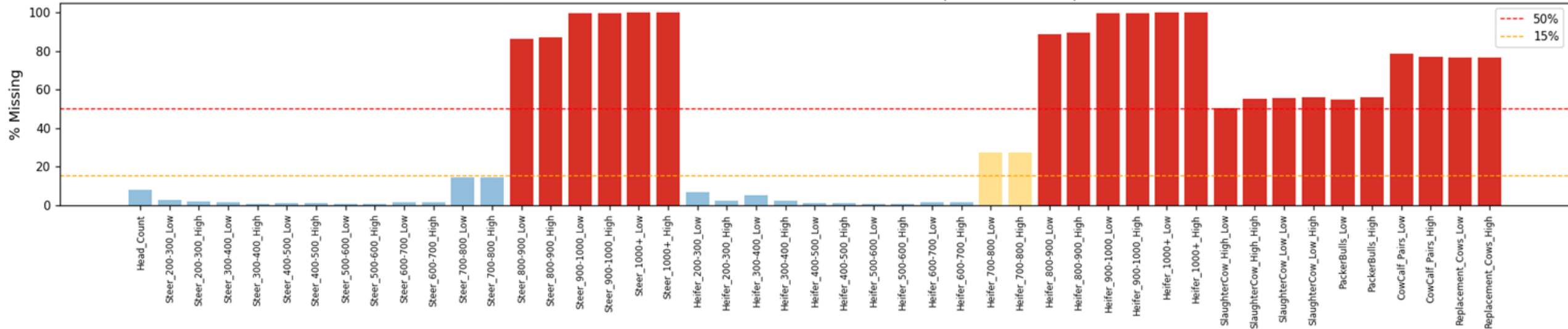
28 barns for clustering

Extended review

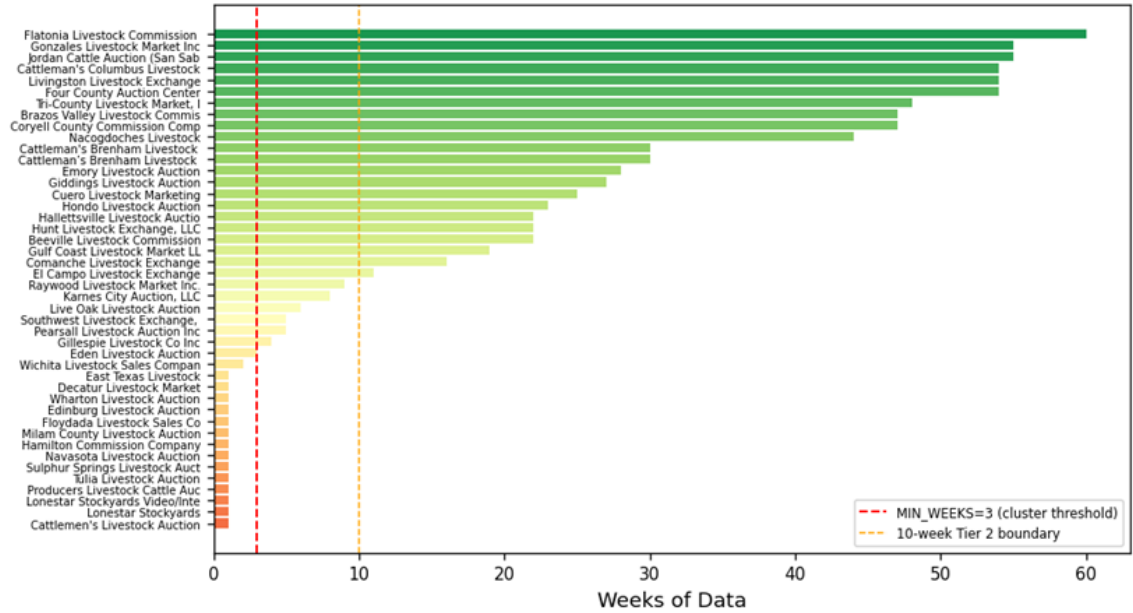
41 barns for anomaly screening

Data Overview

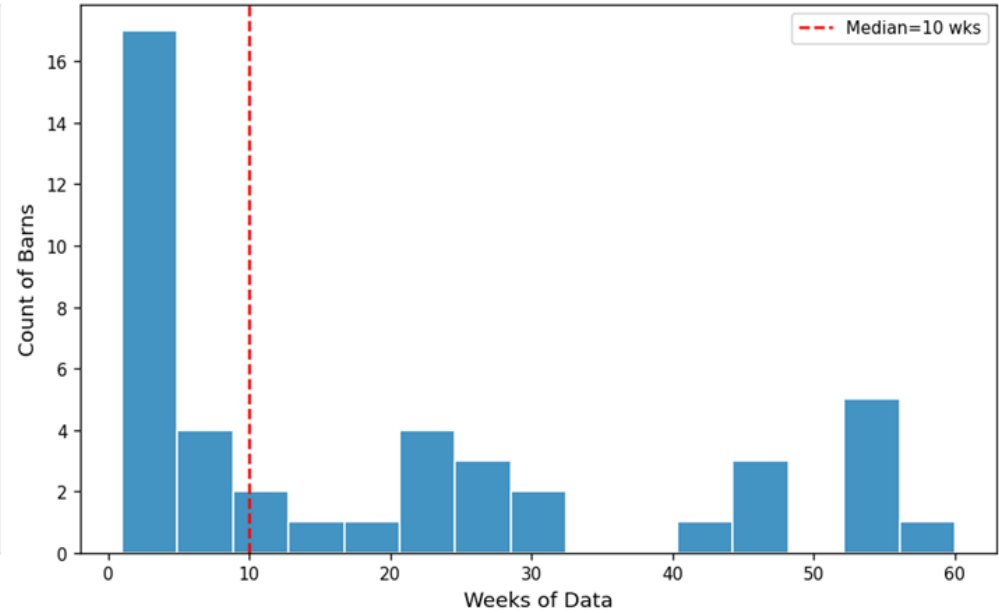
Missing-Value Profile — All Numerical Columns (Red >50% | Amber >15% | Blue <15%)



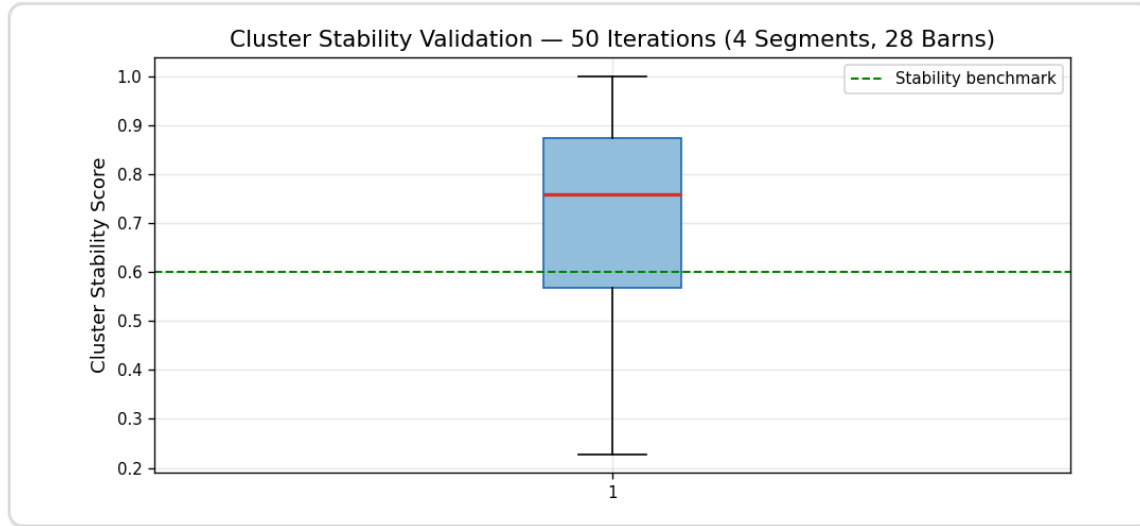
Temporal Coverage per Barn



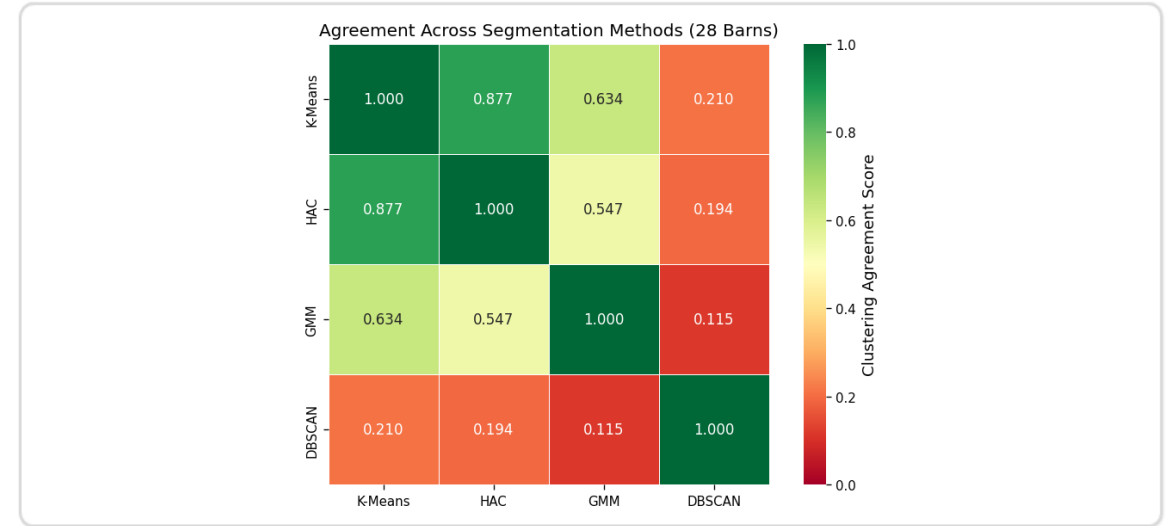
Distribution of Temporal Coverage



Data Analysis



Cluster stability validation across 50 iterations



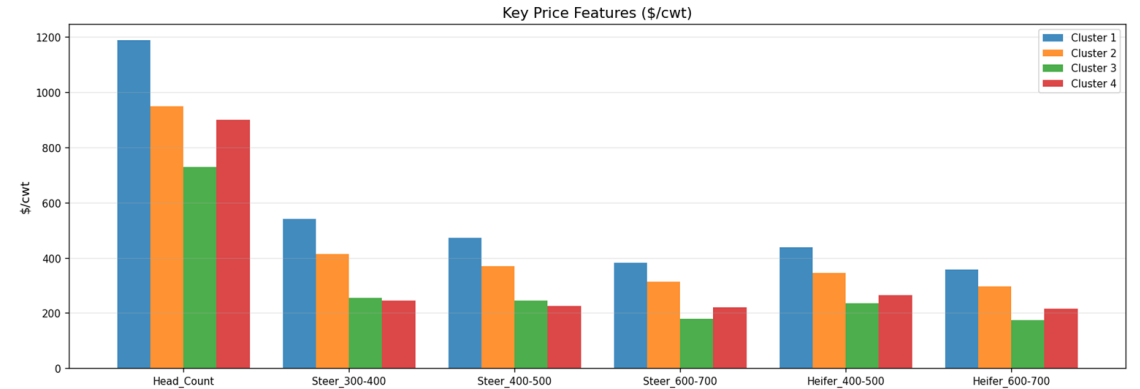
Agreement matrix of different analysis techniques

- Clear structure in the market data
- Distinct market segments identified
- Segments remain consistent across different approaches
- Outliers analyzed separately to avoid distortion

Four Distinct Clusters



Segmented Profiles



Key Centroid Comparison

- C1 - Premium South TX**
highest price + headcount
- C2 - Balanced Mid-Market**
15 barns; benchmark segment
- C3 - Below-Average Niche**
directional, lower-price niche
- C4 - East TX Piney Woods**
stable East Texas cluster

Key Insights

Cluster	Barns	Price Range (\$/cwt)	Characteristics
C1 – Premium South TX Brush Country	7	~\$470–\$540	<i>Highest prices and head count in state. Anchor for premium segment.</i>
C2 – Balanced Mid-Market	15	~\$370–\$413	<i>State price benchmark. Most data-rich cluster (highest confidence).</i>
C3 – Below-Average Niche Markets	2	~\$245–\$256	<i>Data quality caveats; treat as directional.</i>
C4 – East TX Piney Woods	4	~\$226–\$246	<i>Geographically coherent below-average cluster.</i>

Cluster Summary Output

Tier	Weeks of Data	Confidence Level	Usage
Tier 1	≥ 30 weeks	High	<i>Reliable for strategy & benchmarking</i>
Tier 2	10–29 weeks	Moderate	<i>Directional insights</i>
Tier 3	3–9 weeks	Low	<i>Limited use, caution required</i>
Excluded	1–2 weeks	Very Low	<i>Not used in core analysis</i>

Confidence Tiers

What this means:

- Market is segmented
- Value concentrated in **C1 and C2**
- **C2 is the best initial focus**
- C3 and C4 are lower priority

Bottom line:

EDA supports focused penetration over broad coverage.

Market Sizing Funnel

TAM Total Addressable Market | ~180 TX barns (26 observed barns + 60% scale adjustment)

~\$5.9B annual livestock transaction volume

SAM Serviceable Available Market | 60-80 barns in geographic footprint

\$2.7B - \$3.7B midpoint \$3.2B used in modelling

SOM Conservative (5%) ~\$160M | ~4 barns

~\$208K/yr revenue

SOM Base Case (15%) ~\$480M | ~12 barns

~\$624K/yr revenue

SOM Optimistic (25%) ~\$800M | ~20 barns

~\$1.04M/yr revenue

Market Value to CSB Revenue

Payment Rails

~69%

of revenue

Float Income

~13%

of revenue

Embedded Credit

~18%

of revenue

Base case:

\$624K/yr across 12 barns | \$52K per barn | Payment rails leads at 69%

Validated against Lexington real DDA data

Model Validation

Assumption	Verdict
Settlement Structure	✓ Confirmed - 95.3% check-based, exact match
Digital Adoption	✓ Gap = Opportunity - 4.7% today vs 30% target
Revenue lever proportions	✓ Confirmed - <2% shift from projected
Float income	↻ Recalibrated - settlement window tighter than assumed
Per-barn revenue at target	✓ Model holds - within ~14% projection
Compliance risk	✓ Confirmed - exceptions present but <1%

Bottom line: Model structure validated. One recalibration on float. The 4.7% vs 30% adoption gap is the opportunity - closing it is the platform's entire job.

Business Viability

Commercial case is attractive but only for a narrow reachable/monetizable portion of the market.

TAM ~\$5.9B <i>Texas auction value</i>	SAM \$2.7–\$3.7B <i>Reachable footprint</i>
Base Case ~\$624K <i>12 barns / 15% SAM</i>	Upside ~\$1.04M <i>25% SAM / ~20 barns</i>

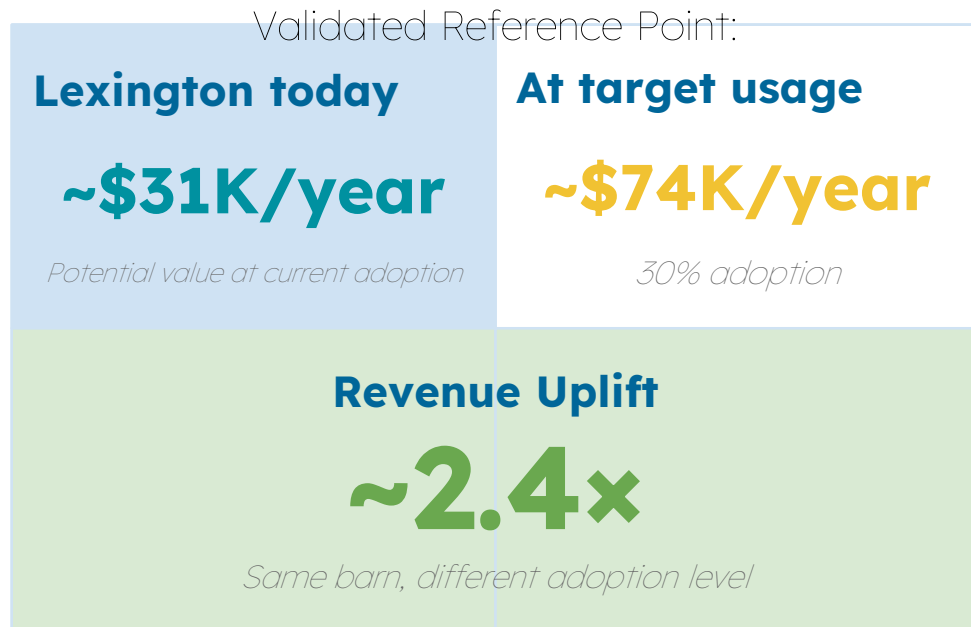
Key economics: ~52K revenue per median barn, with payment rails contributing 69% of total revenue.

Core Value:

Large headline market, but viability depends on converting a narrow, serviceable subset into repeatable revenue.

Economic Value

Barns and users can change their behavior if there is clear value:



Faster Settlement

Reduced Reconciliation Effort

Improved Visibility & Tracking

Fewer Manual Errors / Follow-ups

Compliance Support

Market Adoption

Adoption is the most sensitive variable

Revenue	Barns	Flow	Revenue Lens
5%	~4	\$160M	Early Proof
15%	~12	\$480M	Base Case
25%	~20	\$800M	Upper Bound

Target segment priority

- C2: Lowest-risk entry point
- C1: Offers higher value but requires more trust and relationship

Core Value:

Adoption is relationship driven, not mass marketed.
→ scale, stability, and readiness

Operational Feasibility

Handling non-uniform data and edge cases

- Price Units *\$/cwt / \$/lb / \$/head*
- Incomplete Data
- Data Anomalies

Core Value:

Feasibility is about managing operational complexity
without introducing manual processing

Conclusion

- Market Is Segmented
- Base Case Is Possible
- Value to Barns Must be Obvious
- Adoption Is Relationship Driven

Next Steps

Questions that need Answers

1. How can CSB modernize payment in the context of auction barns?
2. How can CSB connect payment to the sale barn accounting and bookkeeping systems that already exist?
3. How can CSB market modernization to the agro-livestock industry?

A large, weathered bronze sculpture of a longhorn stands prominently in the center of the frame. The sculpture is highly detailed, showing the texture of the animal's hide and the curve of its massive horns. It is set in a park-like environment with a large, leafy tree directly behind it. The ground is a mix of dirt and sparse grass. In the background, a clear blue sky is visible, and a few distant structures and a street lamp can be seen on the right side. The overall scene is bright and sunny.

THANK YOU.

Columbus State Bank



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Texas McCombs
MS Business Analytics
McCombs School of Business